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# An improved 2-pentanone low to high-temperature kinetic model using Bayesian Optimization algorithm

Qianjin Lin, Chun Zou*, Shibo Liu, Yunpeng Wang, Lixin Lu, Chao Peng

*State Key Laboratory of Coal Combustion, Huazhong University of Science and Technology, Wuhan 430074, Hubei, PR China*

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ABSTRACT

The ignition delay times (IDTs) of 2-pentanone (methyl propyl ketone, MPK) were measured at equivalence ratios of 0.5, 1.0, and 1.5, pressures of 1 bar and 5 bar, and temperatures ranging between 1227 and 1571 K. A MPK low to high-temperature model was constructed on the basis of Pieper model and Fenard model. The improvement of the model is that the rate constants of sixteen MPK decomposition and hydrogen abstraction reactions (R1–R16) obtained by the analogy-based method, were globally optimized by the Bayesian Optimization algorithm using the high-temperature IDTs. The optimized model well predicts the laminar flame speeds (measured by Li et al.) and the IDTs and species profiles in low-temperature (measured by Fenard et al.). There is no overfitting during the optimization process. Therefore, the optimization method and the optimized MPK model are reliable. The comparisons of the optimized model with Pieper model and Fenard model were performed by the reaction pathway analysis and sensitivity analysis for the predictions of the MPK profile, the low and high-temperature IDTs and the laminar flame speeds. Because of the difference in the molecular structure between MPK and 2-butanone, remaining the branching ratio among three MPK decompositions (R1–R3) and the ratio among the rate constants of the MPK hydrogen abstraction reactions by OH, H and $CH_3$ unchanged is unnecessary in the development of the MPK model using the analogy-based method, while the branching ratio of the hydrogen abstraction reactions at carbon 1, 3, and 5 remains unchanged. The competition among the decomposition and hydrogen abstraction reactions of MPK is intricate and crucial to the low and high-temperature oxidation. It is necessary to globally optimize the rate constants of R1-R16 based on the multidimensional experimental results, such as low and high-temperature IDTs, laminar flame speeds, and species profiles.



## 1. Introduction

Declining fossil fuel resources and the general desire for a reduction of greenhouse gas emissions have led to a rising interest in biofuels [1,2]. 2-Pentanone (methyl propyl ketone, MPK) is a type of biofuel, which can be converted from cellulosic biomass [3–5]. Moreover, MPK could be suitable for engine applications due to the high energy density, impressive knock resistance, and low soot emissions [6–8]. Therefore, the experimental and modeling studies on MPK have attracted the interest of researchers.

The experimental studies of MPK reported in previous literature are summarized in Table 1. Lam et al. [9] measured the ignition delay times (IDTs) of MPK at 2.65 bar under the $O_2$/Ar atmosphere. Subsequently, Minwegen et al. [10] measured the IDTs of MPK at 20 bar and 40 bar under the $O_2/N_2$ atmosphere. Pieper et al. [7] measured the flame species profiles of MPK at 40 mbar under the $O_2$/Ar atmosphere and Li et al. [11] measured the laminar flame speeds (LFSs) of MPK at 1–10 atm under the $O_2/N_2$ atmosphere. Recently, Fenard et al. [12] measured the IDTs of MPK in the low temperature range of 650–950 K at 20 bar and 40 bar and the species profiles in a plug flow reactor (PFR) in the temperature range of 800–1050 K. Ignition delay time (IDT) is a crucial parameter for the combustion characteristics of fuels and kinetic model validation [13–16], previous IDT measurements mainly focus on the stoichiometric and high pressure conditions. To our knowledge, the IDTs of MPK at fuel-lean and fuel-rich conditions at relatively low pressures have not been measured.

Pieper et al. [7] carried out the pioneering and excellent work on the MPK high-temperature model (named as Pieper model), in which the MPK sub-model was constructed by the analogy-based method. Pieper model has been validated by the species profiles measured by themselves and the literature IDTs [9,10]. Subsequently, Fenard et al. [12] developed a new MPK model (named as Fenard model) for the low and high-temperature MPK oxidation using the analogy-based method. Fenard model well

* Corresponding author.
*E-mail address:* zouchun@hust.edu.cn (C. Zou).

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**Table 1**
Experimental IDTs, species profiles, and laminar flame speeds reported in previous literature for MPK.

| Type of experiment | Experimental settings | | | | Reference |
|---|---|---|---|---|---|
| | $T$ | $\phi$ | $P$ | Diluent | |
| Shock tube | 1300–1500 K | 1 | 2.65 bar | Ar | Lam et al. [9] |
| | 1020–1220 K | 1 | 20,40 bar | $N_2$ | Minwegen et al. [10] |
| Rapid compression machine | 650–950 K | 1 | 20,40 bar | $N_2$ | Fenard et al. [12] |
| Species profiles | 400–2200 K | 1.6 | 40 mbar | Ar | Pieper et al. [7] |
| | 800–1050 K | 0.8 | 970 mbar | Ar | Fenard et al. [12] |
| Spherical bomb | 423 K | 0.6–1.5 | 1 atm | $N_2$ | Li et al. [11] |
| | | 0.6–1.5 | 2 atm | | |
| | | 0.6–1.1 | 5 atm | | |
| | | 0.6–1.0 | 10 atm | | |
| | 393 K | 0.7–1.5 | 1 atm | | |
| Heat flux burner | 393 K | 0.7–1.5 | 1 atm | | |

predicts the low-temperature experimental data and literature high-temperature IDT data [9,10]. However, the species profiles predicted by Pieper model significantly deviate from the measured results of [12] while Fenard model underpredicts the laminar flame speeds measured by Li et al. [11]. In the present study, an improved MPK low to high-temperature oxidation model was proposed based on Pieper model and Fenard model.

Some decomposition and hydrogen abstraction reactions of MPK were obtained by the analogy method due to the difficulty in the theoretical calculations and experimental measurements. In Pieper model, the rate constants of the MPK decomposition reactions, R1–R3 and R4 were obtained by analogy with 2-butanone and pentane [17,18], respectively, which were increased by a factor of four to match the IDTs measured by Lam et al. [9] and Minwegen et al. [10]. In Fenard model, the rate constants of R1–R3 were also obtained by analogy with 2-butanone (R1–R3) [17], while the rate constants of R4 are the same as R3 and increased by a factor of two to match the IDTs from Refs. [9,10]. However, the rate constants of R4 in Pieper model and Fenard model are higher than that of R3 while the bond dissociation energy (BDE) of R4 (88.1 kcal/mol) is higher than that of R3 (82.2 kcal/mol). In other words, R3 is underestimated in both Pieper model and Fenard model.

| | |
|---|---|
| R1 | MPK(+M)<=>$CH_3$+$NC_3H_7CO$(+M) |
| R2 | MPK(+M)<=>$NC_3H7$+$CH_3CO$(+M) |
| R3 | MPK(+M)<=>$C_2H_5$+$CH_3COCH_2$(+M) |
| R4 | MPK<=>$CH_3$+$CH_2CH_2COCH_3$ |
| R5–R8 | MPK + OH <=> MPK-X+$H_2O$ |
| R9–R12 | MPK + H <=> MPK-X+$H_2$ |
| R13–R16 | MPK + $CH_3$ <=> MPK-X + $CH_4$ |

MPK-X represents the hydrogen abstraction products at carbon X, $X$ = 1, 3, 4, and 5

The fuel molecule hydrogen abstraction reactions show high sensitivity to the IDT [19,20]. In Pieper model, the rate constants of MPK hydrogen abstraction reactions by OH (R5–R8), H (R9–R12), and $CH_3$ (R13–R16) radicals were obtained by analogy with 2-butanone at carbons 1, 3, and 4 and pentane at carbon 5, respectively, while these reactions in Fenard model were obtained by analogy with 2-butanone at carbons 1, 3, and 5 and pentane at carbon 4. As depicted in Fig. 1, although the carbon−hydrogen (C–H) BDE of MPK at carbon 4 is close to that of pentane, the gap between the two as well as the existence of carbonyl could result in the significant deviation in the rate constants of R7, R11 and R15 (at carbon 4) from the corresponding reactions of pentane.

The study of You and co-workers [23] shows that the reaction rate constants obtained by the analogy-based method are not necessary to be the same as the original rate constants. Therefore, the rate constants of R1–R16 obtained by the analogy-based method could be adjusted to improve the prediction performance. Even more important, in the MPK low to high-temperature model, R1–R16 compete with each other for MPK molecule, which are crucial to the low-temperature and high-temperature oxidation of MPK. As a result, the reaction rate constants of R1–R16 are in need of global optimization based on the experimental data listed in Table 1. In our previous work [24], the rate constants of eight 3-pentanone (DEK) related reactions obtained by the analogy-based method were globally optimized by the Bayesian Optimization algorithm (BOA), and the optimized DEK model predicted well the IDTs, laminar flame speeds, and species profiles. BOA is a strong global optimization strategy, and ideally suitable for expensive cost tasks [25]. Therefore, BOA was employed in the present study to implement the optimization.

In the present study, IDTs of MPK were measured at equivalence ratios of 0.5, 1.0, and 1.5, pressures of 1 bar and 5 bar, and temperatures ranging from 1227 to 1571 K. A MPK low to high-temperature submodel was constructed on the basis of Pieper model and Fenard model and optimized by the BOA using the high-temperature IDTs measured by Lam et al. [9] and Minwegen et al. [10] and the present study. Subsequently, the low-temperature IDTs, the species profiles, and the flame speeds listed in Table 1 were utilized to validate the MPK low to high-temperature submdel. The comparisons of the optimized MPK model to Pieper model, and Fenard model were performed. The principles of the optimization of the decomposition and hydrogen abstraction reactions of MPK were discussed.

## 2. Methodology

### 2.1. Shock tube facility and the definition of IDT

The shock tube used in present study has been reported in detail in our previous work [26–28] and only a brief description is provided here. The shock tube consists of a 4-m driver section and an 8-m driven section with an inner diameter of 100 mm. Two different thicknesses of polyester terephthalate diaphragms were used to obtain the IDT at two different test pressures (about 1 and 5 bar). Table 2 lists the experimental conditions investigated in this study. The purities of oxygen, argon, MPK (Aladdin) were respectively 99.999%, 99.999%, and 99%. The partial pressure of MPK was below 50% of its saturated vapor pressure (3.56 kPa at 293 K [29]) to ensure that MPK was gas-phase. Prior to each experiment, the shock tube was evacuated to less than $1 \times 10^{-6}$ bar.

The IDT was defined as the interval of the reflected wave and the onset of the ignition. In this study, the onset of ignition was calculated by linearly extrapolating the time at which the steepest increase in OH* radical emission occurs to the zero baseline [13,26]. Figure 2 depicts a typical signal trace obtained in the present study and the method of determination of the IDT. The

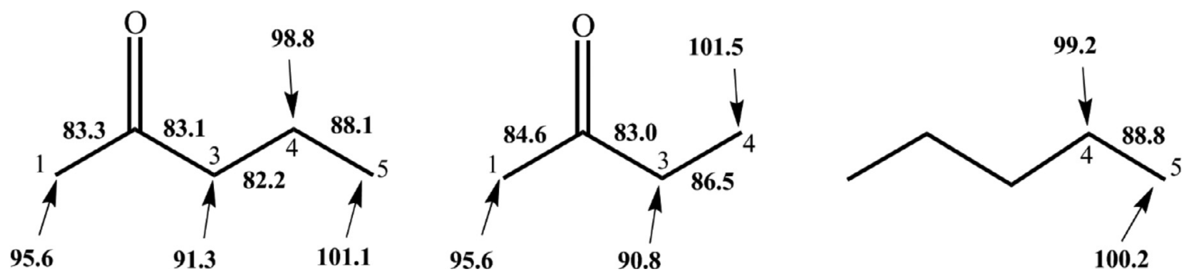


**Fig. 1.** C–H and C–C BDEs (kcal $mol^{-1}$) of MPK [7], 2-butanone [21], and pentane [22].

**Table 2**
Experimental conditions in the shock tube.

| Mix | Pressure (bar) | $\varphi$ | MPK (%) | $O_2$ (%) | Ar (%) |
|---|---|---|---|---|---|
| 1 | 1 | 0.5 | 1 | 14 | 85 |
| 2 | 1 | 1 | 1 | 7 | 92 |
| 3 | | | | | |
| 3 | 1 | 1.5 | 1.5 | 7 | 91.5 |
| 4 | 5 | 0.5 | 1 | 14 | 85 |
| 5 | 5 | 1 | 1 | 7 | 92 |
| 6 | 5 | 1.5 | 1.5 | 7 | 91.5 |

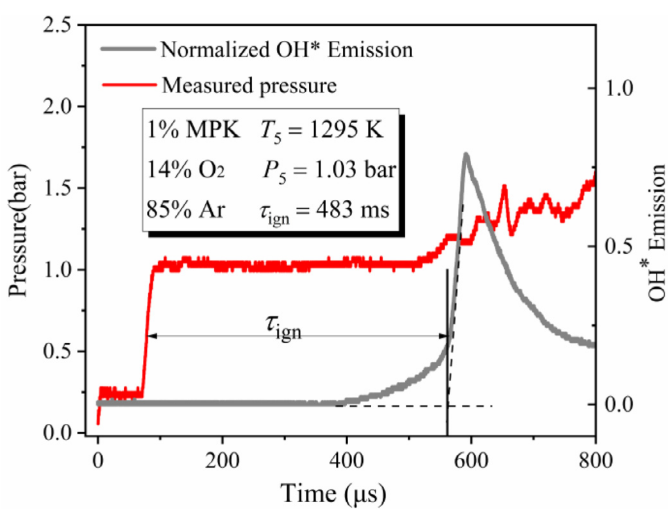


**Fig. 2.** Definition of IDT for Mixture 4 at $T = 1295$ K and $P = 1.03$ bar.

temperature (T5) and pressure (P5) behind the reflected shock wave were calculated using Gaseq [30], based on the measured shock wave velocity obtained by piezoelectric pressure transducers. The uncertainty of measurement was mainly from the uncertainty of T5, and the uncertainty of T5 was evaluated by using the same method as in Ref [26]. As a result, the uncertainty in the IDT measurements was less than 20%.

### *2.2. Simulation methods*

The IDTs were simulated with homogeneous, adiabatic, and constrained volume assumptions. The calculations of the IDT were conducted in Cantera 2.4.0 [31] to integrate the Bayesian Optimization algorithm. The laminar flame speeds and the flame species profiles were calculated using CHEMKIN-PRO software [32]. Flame speed calculations were performed using mixture-averaged transport properties and considering Soret effects. A maximum of 1500 grid points is adopted, and CURV and GRAD values of 0.02 were attained, ensuring that the final solution was independent of the number of grid points. The flame species profiles were simulated using the PREMIX module in CHEMKIN-PRO [32], in which the measured temperature profile was used as the input parameter. The species profiles measured in the PFR were simulated using the Plug Flow Reactor module in CHEMKIN-PRO [32], in which the calculated temperature profile was also used as the input parameter.

### *2.3. Model optimization method*

The BOA has been reported in detail in our previous work [24], and only a brief description is presented here. The typical BOA [25] works by using the known observations calculated by an objective function to fit a Gaussian process model (a probabilistic surrogate model) and using an acquisition function to get the posterior sample location. The objective function determines the aim of the optimization, which is defined as:

$$f\left(\overrightarrow{X_i}\right) = \frac{1}{N}\sum_{n=1}^{N}\left|\frac{\tau_k - \tau_0}{\tau_0}\right| \times \ 100\% \tag{1}$$

Where $f(\overrightarrow{X_i})$ denotes the value of the objective function at the $i$th sampling point; $N$ denotes the data points in the dataset; $\tau_k$ and $\tau_0$ are the calculated and measured IDTs, respectively.

The next optimum position, $X_{n+1}$, is determined by the acquisition function.

Finally, a terminal condition is defined in Eq. (2) to stop the optimization loop:

$$\text{Terminal condition} = \begin{cases} Objective\ value \ \le 20\% \\ \quad |f_{i-10} - f_i| \le 1\% \end{cases} \tag{2}$$

Where $f_{i-10}$ and $f_i$ denote the objective function value of the $(i-10)$th sampling point and the $i$th sampling point, respectively.

## 3. MPK low to high-temperature kinetic model

A MPK low to high-temperature submodel was constructed based on Pieper model and Fenard model. Pieper MPK submodel was adopted as the high-temperature submodel of MPK, in which the rate constants of R1–R4 were obtained by analogy with 2-butanone (R1–R3) and pentane (R4) [17,18] respectively, and the rate constants of the R5–16 were obtained by analogy with 2-butanone at carbons 1, 3, and 5 and pentane at carbon 4, which were also adopted in Fenard model. The low-temperature submodel in Fenard model was adapted. The rate constants of R17 (MPK-$OOH_3$-R1+$O_2$<=>MPK-$OOH_3$-OO1) and R18 (MPK-$OOH_4$-R1+$O_2$<=>MPKOO$H_4$-OO1) are the same as those of the first $O_2$ addition reactions in the present MPK low to high-temperature model. The core mechanism of the present model is AramcoMech 2.0 with a 2-butanone submechanism [33] and a small-aromatic submechanism [34]. The detailed MPK low to high-temperature kinetic model is provided in Supplemental Materials.

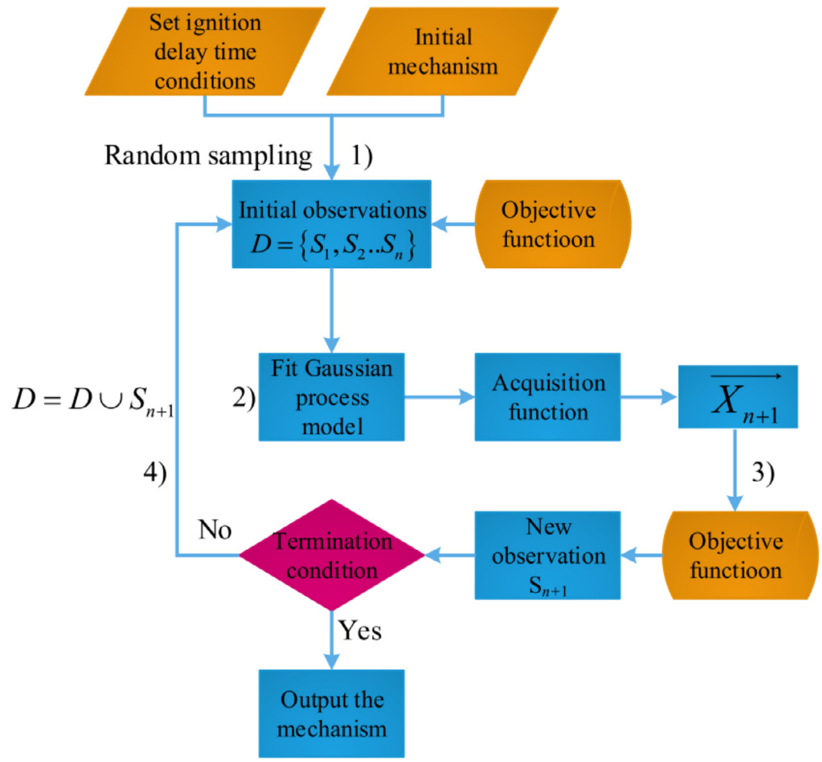


**Fig. 3.** Implementation process of model optimization using the BOA.

## 4. MPK model optimization using BOA

Theoretically, all reaction rate coefficients of the 16 elementary reactions should be optimized. However, most researchers prefer to modify the pre-exponential factors [35–39] because the pre-exponential factors have intuitive effects on the reaction process, whereas the modification of other coefficients may cause reactions not to proceed [40]. Therefore, the pre-exponential factors of the 16 elementary reactions were selected for the optimization in this work. The adjustment factor is defined as the ratio of the modified pre-exponential factor to the original pre-exponential factor, and the value ranges of the adjustment factors are generally equal to the uncertainty factors of the reactions. To maintain the branching ratios of MPK hydrogen abstraction reactions by OH radical, as well as by H and $CH_3$ radicals, at carbons 1, 3, and 5 unchanged during the optimization process, the adjustment factors ($x$) of the MPK hydrogen abstraction reactions by the identical radical are the same [24]. Therefore, the number of the adjustment factors is 10.

Pieper et al. increased the rate constants of MPK decomposition reactions (R1–R4) by a factor of four in the MPK kinetic model due to the high collision probability of MPK compared with 2-butanone. Therefore, the uncertainty factors of R1, R2 and R4 are 4, and the ranges of adjustment factors of the three reactions are from 1/4 to 4 based on the original analogy [17]. The BDE of the third C–C bond in MPK is 4.3 kcal/mol lower than that in 2-butanone [7,21]. The difference of BDE of the third C–C bond between MPK and 2-butanone is three times as many as that of the first C–C bond, thus, the uncertainty factor of R3 are 12 and the range of the adjustment factors of R3 is from 1/12 to 12. The uncertainty factors of the rate constants of 2-butanone hydrogen abstraction reactions by OH, H, and CH3 at sites 1, 3, and 5 are determined as 3 by Zhou et al. [41] and Kopp et al. [42]. Therefore, the uncertainty factors of R5–R16 are 3, and the ranges of adjustment factors of R5–R16 are from 1/3 to 3. Moreover, the collision limits of the optimized 16 reactions were evaluated using the method proposed by Chen et al. [43] and the results were provided in the Supplementary material. From Fig. S4, the rate coefficients of R1-R16 have not violate its collision limit over a temperature range of 300–2500 K.

The MPK low and high-temperature model was optimized by the high-temperature IDTs. The LFSs, low-temperature IDTs, and species profiles were used for validation.

Figure 3 illustrates the implementation process using the BOA, which is summarized as four steps:

The optimization object is the adjustment vector $S$ including 10 adjustment factors, $x_{i,}$ $i = 1, 2, \ldots, 10$. In the uncertainty ranges of each adjustment factor, 200 initial sampling points are randomly produced ($n = 200$), and the observations $D \in \{S_1, S_2 \ldots S_{200}\}$ were obtained using Eq. (1).

Using the observations, $D$, the Gaussian process model could be established. Then, the acquisition function is employed to get the next potential optimum location, $(\overrightarrow{X_{n+1}})$, which is the potential optimum model.

The objective function is employed to calculate the new observation, $S_{n+1}$, for the $(\overrightarrow{X_{n+1}})$.

The final optimized model would output when the terminal condition shown in Eq. (2) is satisfied. Otherwise, the observation space, $D$, would update by the new observation, $S_{n+1}$, and Steps 2)–4) would repeated iteration by iteration.

At the 160th iteration, the terminal condition is satisfied, as shown in Fig. 4. The optimized pre-exponential factors of the 16 reactions are provided in Table 3.

Figure 5 depicts the simulated results of the optimized model and the model without optimization for the IDTs measured in this work. It can be seen that the optimized model well reproduces the IDTs at all conditions within the 20% error limit, while the model without optimization significantly overpredicts the experimental data at all conditions. Obviously, the optimization significantly improves the performance of the MPK model.

**Table 3.**
Optimized reactions in the present study (the reaction rate constants are expressed as $k = AT^n exp(-E_a/RT)$ cm$^3$ mol$^{-1}$ s$^{-1}$, with $E_a$ in cal mol$^{-1}$).

| Number | Reactions | $A$ | $A_{modify}$ | $A_{modify}/A$ |
|---|---|---|---|---|
| R1 | MPK(+M)<=>$CH_3$+$NC_3H_7CO$(+M) | 3.3E+27 | 1.34E28 | 3.9 |
| | Low pressure limit | 1.16E+72 | 4.70E72 | 3.9 |
| R2 | MPK(+M)<=>$NC_3H_7$+$CH_3CO$(+M) | 3.5E+27 | 1.40E28 | 1.0 |
| | Low pressure limit | 2.6E+78 | 1.03E80 | 1.0 |
| R3 | MPK(+M)<=>$C_2H_5$+$CH_3COCH_2$(+M) | 4.8E+24 | 2.94E25 | 6.0 |
| | Low pressure limit | 5.8E+69 | 3.5E70 | 6.0 |
| R4 | MPK<=>$CH_3$+$CH_2CH_2COCH_3$ | 3.5E+21 | 3.5E21 | 1.1 |
| R5 | MPK + OH <=> MPK-R1 + $H_2O$ | 3.99E2 | 2.45E2 | 0.45 |
| R6 | MPK + OH <=> MPK-R3 + $H_2O$ | 2.36E2 | 1.06E2 | 0.45 |
| R7 | MPK + OH <=> MPK-R4 + $H_2O$ | 7.05E9 | 2.32E9 | 0.34 |
| R8 | MPK + OH <=> MPK-R5 + $H_2O$ | 1.35 | 0.61 | 0.45 |
| R9 | MPK + H <=> MPK-R1 + $H_2$ | 7.38E4 | 2.51E4 | 0.34 |
| R10 | MPK + H <=> MPK-R3 + $H_2$ | 4.94E4 | 1.65E4 | 0.34 |
| R11 | MPK + H <=> MPK-R4 + $H_2$ | 1.30E6 | 0.4E6 | 0.33 |
| R12 | MPK + H <=> MPK-R5 + $H_2$ | 7.79E5 | 2.6E5 | 0.34 |
| R13 | MPK + $CH_3$ <=> MPK-R1 + $CH_4$ | 8.66E-1 | 8.66E-1 | 1.0 |
| R14 | MPK + $CH_3$ <=> MPK-R3 + $CH_4$ | 1.63 | 1.63 | 1.0 |
| R15 | MPK + $CH_3$ <=> MPK-R4 + $CH_4$ | 8.40E4 | 8.40E4 | 1.0 |
| R16 | MPK + $CH_3$ <=> MPK-R5 + $CH_4$ | 2.21 | 2.21 | 1.0 |

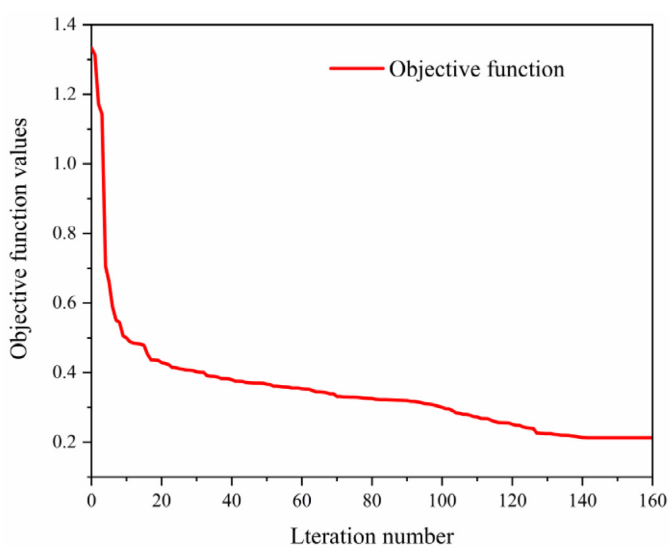


**Fig. 4.** Evolution of the objective function with the literation.

## 5. Results and discussion

### 5.1. Optimized MPK model validation

Figure 6 shows the predicted results of the optimized MPK model, Pieper model and Fenard model for the IDTs obtained in this work. From Fig. 6, the optimized MPK model well predicts the experimental IDTs within the 20% error limit, while the Pieper model overpredicts the IDTs at conditions of $\varphi = 0.5$ and Fenard model overpredicts the IDTs at all conditions.

Figure 7 displays the predicted results of the optimized MPK model, Pieper model and Fenard model for the IDTs reported in literature [9,10,12]. From Fig. 7(a), the optimized model predicts well the high temperature IDTs at 2.65 bar while Pieper model and Fenard model underestimates and overestimates the results at temperatures around 1350 K, respectively. From Fig. 7(b), the optimized model well reproduces the IDTs at 20 bar and 40 bar. Fenard model underpredicts the IDTs at the temperature range of 1080–1200 K and pressure of 40 bar. For the negative temperature coefficient (NTC), the optimized model predicts well the NTC behaviors at 20 and 40 bar. Fenard model overpredicts the low temperature IDTs at the conditions of $P = 20$ bar and temperatures around 700 K while Pieper model fails to predict (NTC) behavior, as shown in Fig. 7(c).

Figure 8 depicts the predicted results of the optimized MPK model, Pieper model and Fenard model for the LFSs measured by Li et al. [11]. From Fig. 8, the optimized MIPK model shows the best agreement with the experimental data, while Fenard model underpredicts the LFSs at pressures of 2–10 atm and Pieper model underpredicts the LFSs at fuel-rich conditions. Fenard model predicts almost all low temperature results within the given experimental uncertainty except at the conditions of $P = 20$ bar and temperatures around 700 K while Pieper model fails to predict (NTC) behavior due to the lack of the low-temperature mechanism of MPK, as shown in Fig. 7(c).

Figure 9 shows the predicted results of the optimized MPK model, Pieper model and Fenard model for PFR species profiles measured by Fenard et al. [12] during the low temperature combustion. From Fig. 9, the optimized model and Fenard model well reproduce the experimental data while Pieper model fails to capture the decay of MPK and oxygen molecule or the rise of $CO_2$ and $H_2O$.

In the Supplemental material, the predicted results of the optimized MPK model, Fenard model and Pieper model were compared with the experimental species profiles from Pieper et al. [7] and Fenard et al. [12].

The rate constants of R1–R16 were optimized using the high temperature IDTs measured by Lam et al. [9] and Minwegen et al. [10] and the present IDTs, and the optimized MPK model predicts well the LFSs, species profiles, and the low temperature IDTs listed in Table 1. These indicate that there is no overfitting during the optimization process. Therefore, the optimization method and optimized MPK model are reliable.

### 5.2. Model comparisons for PFR species profiles prediction

For MPK profiles measured in PFR and $T = 950$ K, the optimized model well predicts the MPK concentration while Pieper model underestimates the MPK concentration by ~70%. Fig. 10 shows the MPK sensitivity analysis results using the optimized model and Pieper model at $T = 950$ K. From Fig. 10, R2 (MPK(+M)<=>$C_2H_5$+$CH_3COCH_2$(+M)), the fuel decomposition reaction, makes the largest positive contribution to the MPK consumption in two models. The rate constants of R2 was increased by a factor of 4 in Pieper model and was restored in the optimized model. As displayed in Fig. 11, 5.0% fuel is consumed by R2 in the optimized model, while the corresponding values are 18.9%

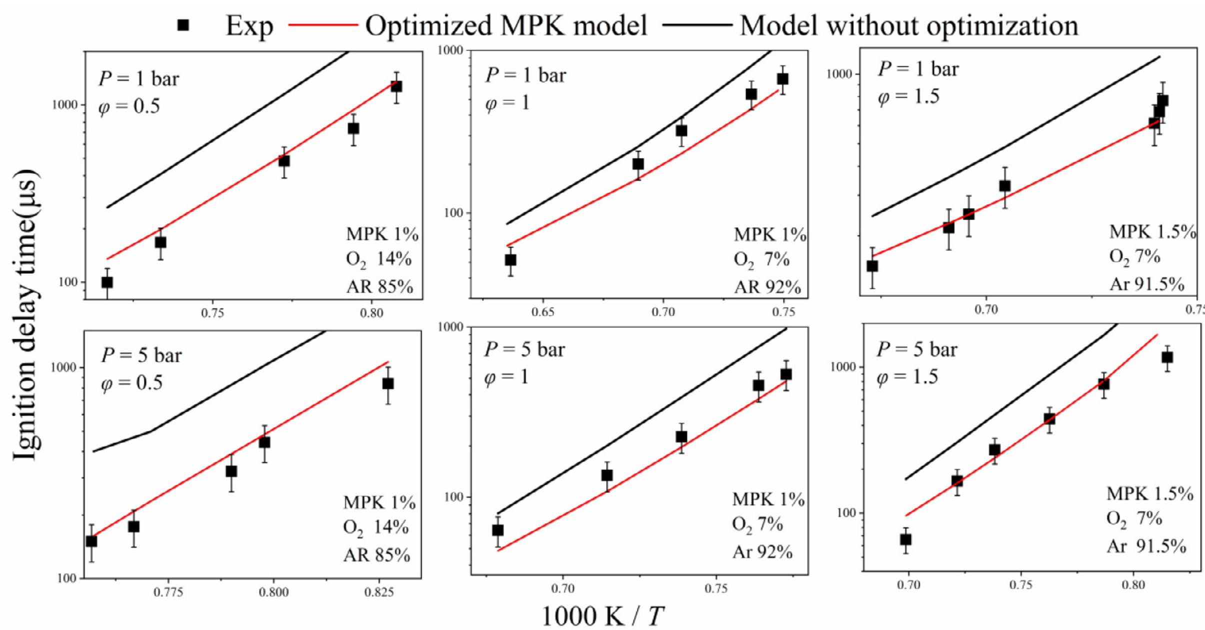


**Fig. 5.** Measured IDTs and simulated results of the optimized model and the model without optimization.

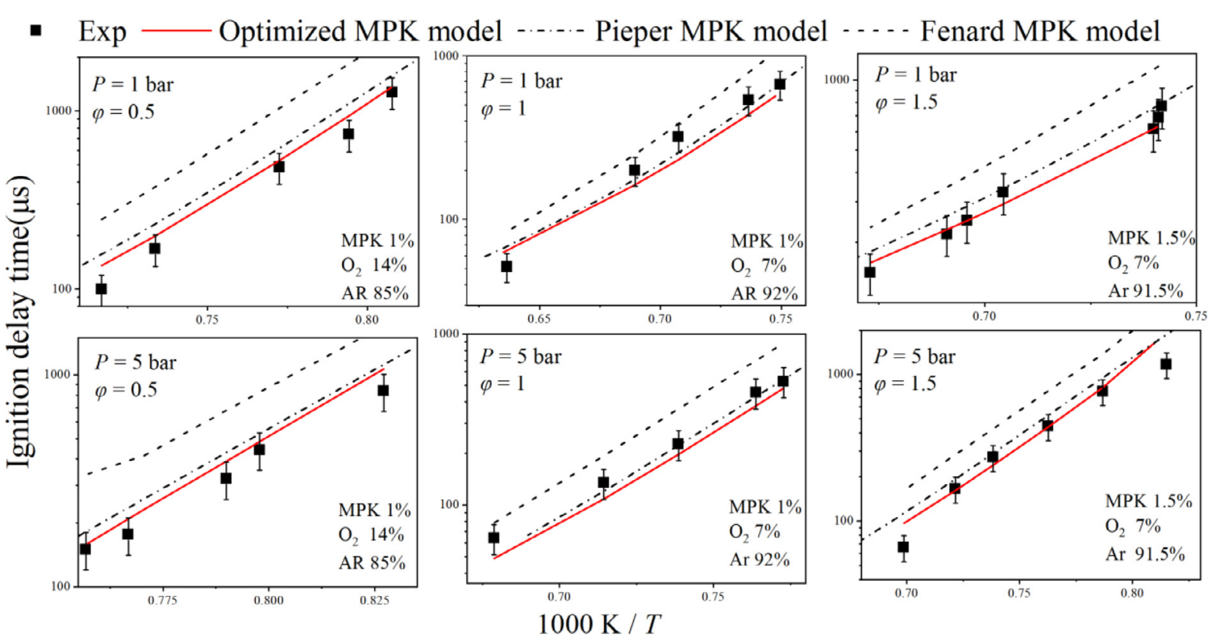


**Fig. 6.** Measured IDTs and simulated results of the optimized model, Pieper model, and Fenard model.

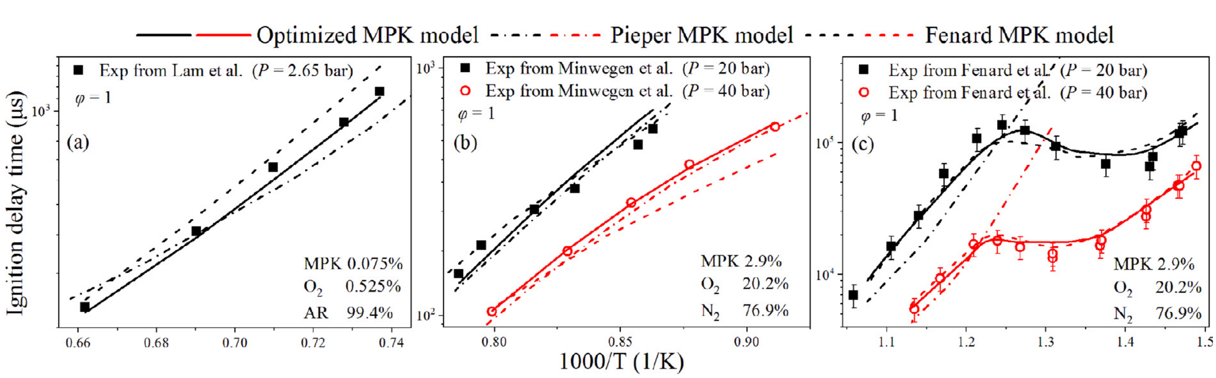


**Fig. 7.** Experimental IDTs reported in literature [9,10,12] and simulated results of the optimized model, Pieper model, and Fenard model.

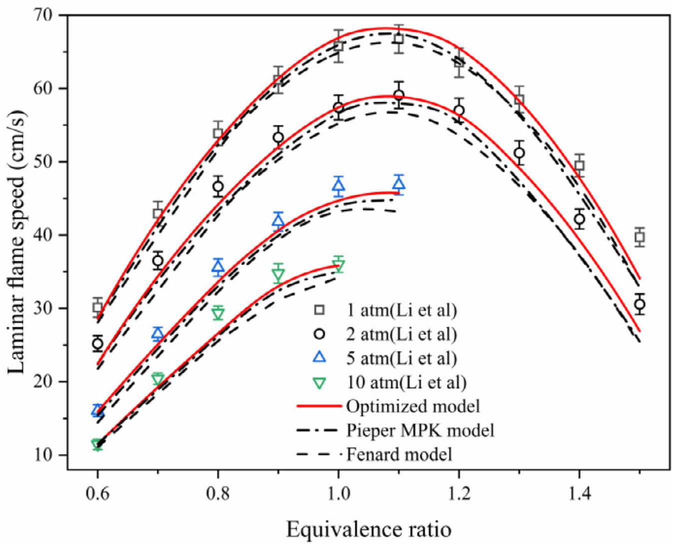


**Fig. 8.** Experimental LFSs from Li et al. [11] and simulated results of the optimized model, Pieper model, and Fenard model.

in Pieper model. Consequently, the overestimation of the R2 is the main reason for the significant underestimation of the MPK concentration in Pieper model.

### 5.3. Model comparisons for IDT prediction at 1250 K and 700 K

At the condition of $\phi = 0.5$, $P = 5$ bar, and $T = 1250$ K, the optimized model well predicts the experimental IDTs, while Fenard model and Pieper model overpredict the experimental IDTs by ~200% and ~30% respectively. The temperature sensitivity analysis at the ignition point, which is approximately equal to the sensitivity of the IDT [44], were conducted using the

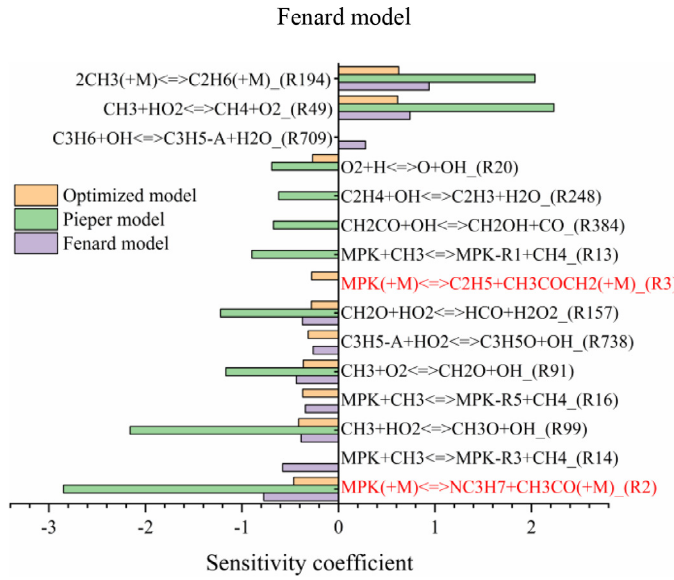


**Fig. 10.** MPK sensitivity analysis at 950 K using the optimized model, Pieper model and Fenard model.

optimized model, Pieper model and Fenard model and the results are shown in Fig. 12. The positive value means promoting effects to the ignition, while the negative value means suppressing effects to ignition. For the optimized model, R2 and R3 (MPK(+M)<=>$CH_3$+$NC_3H_7CO$(+M)), rank third and fifth place among the ignition promoting reactions, respectively. For Fenard model, R2 ranks the first place among the ignition promoting reactions while R3 has almost no influence on the ignition. Figure 13 plots the reaction pathway analysis results using the optimized model, Pieper model and Fenard model. From Fig. 13, 8.0% fuel is consumed by R3 in the optimized model while the corresponding

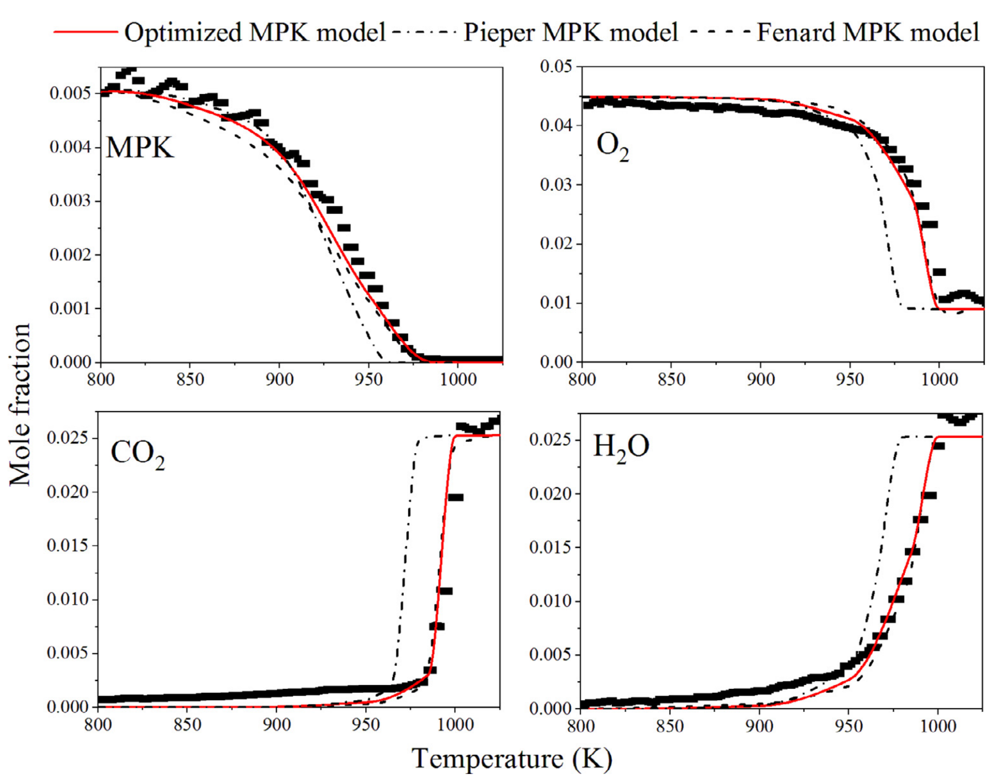


**Fig. 9.** Experimental species profiles measured by Fenard et al. [12] and simulated results of the optimized model, Pieper model, and Fenard model.

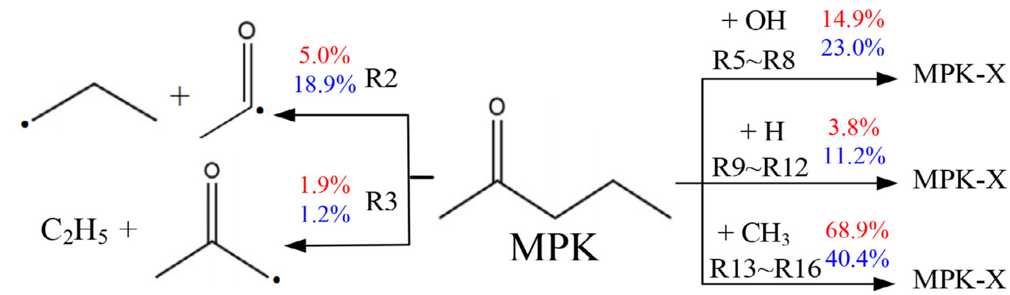


**Fig. 11.** Reaction flux analysis for MPK oxidation in PFR at 950 K using the optimized model (red) and Pieper model (blue). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

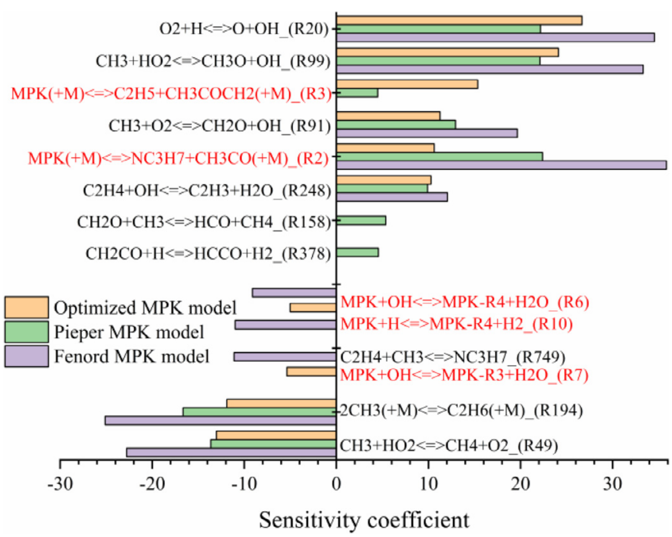


**Fig. 12.** Temperature sensitivity analysis using the optimized model, Pieper model, and Fenard model at the ignition point; $\phi$ = 0.5, 1250 K and 5 bar.

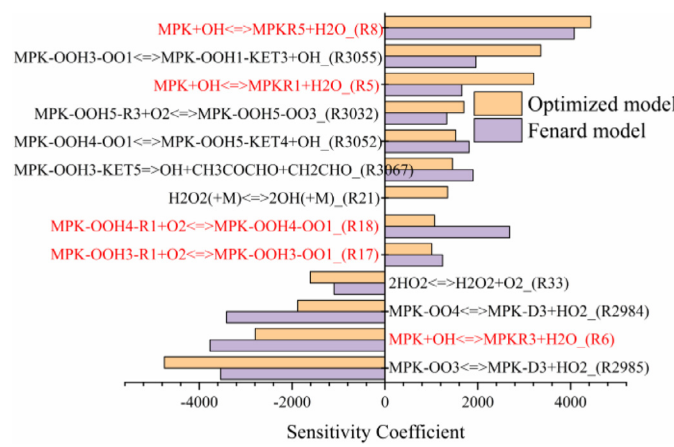


**Fig. 14.** Temperature sensitivity analysis using the optimized model and Fenard model at the ignition point; $\phi$ = 1, 700 K, and 20 bar.

value is 0.8% in Fenard model. Therefore, the underestimation of R3 is one reason for the significant overprediction of the IDTs in Fenard model. In addition, R6, R7 and R11 show obvious inhabiting effects to the ignition in the optimized model and Fenard model. From Fig. 13, 12.4% fuel is consumed by R6, R7, and R11 in the optimized model while the corresponding value is 18.1% in Fenard model. Consequently, the overestimation of R6, R7, and R11 is another reason for the significant overprediction of the IDTs in Fenard model.

From Fig. 13, 3.3% fuel is consumed by R3 in Pieper model, which shows that the underestimation of R3 is also the reason for the overprediction of the IDTs in Pieper model. The amount of MPK consumed by the H-atom abstractions by OH and H in the optimized model is 22.8% and 107.7% more than those in the Pieper model, respectively. The underestimation of the H-abstraction reactions by OH and H subsequently results in the absence of the H-abstraction reactions in Fig. 12.

At the condition of $P$ = 20 bar and $T$ = 700 K, the optimized model well predicts the experimental IDTs within the 20% uncertainty, while Fenard model overpredicts the experimental IDTs by 28%. The optimized model and Fenard model are used to perform the temperature sensitivity analysis and the results are shown in Fig. 14. For the optimized model, R8 and R5 rank first and third place among the ignition promoting reactions, respectively. For Fenard model, R8 and R18 rank first and second place among the ignition promoting reactions, respectively. The reaction pathway of MPK in the optimized model and Fenard model is shown in Fig. 16. The ratio of MPK consumed by R8 is 15.7% and 13.4% in the optimized model and Fenard model respectively, thus, the underestimation of R8 is one reason for the overprediction of the IDTs in Fenard model. From Fig. 15, the MPK-R4 produced by R7 is the

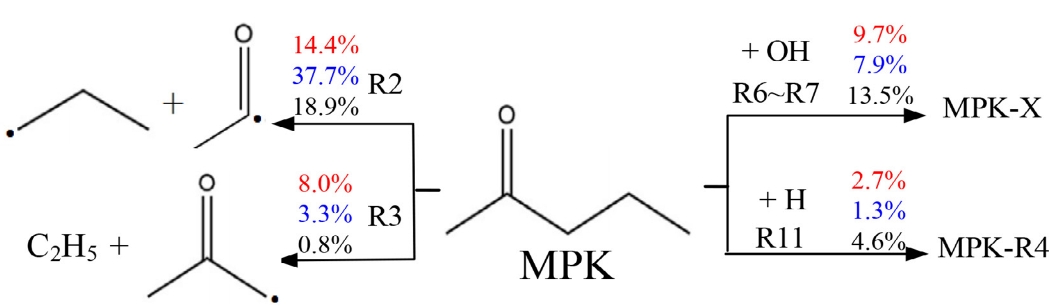


**Fig. 13.** Reaction flux analysis for MPK using the optimized model (red), Pieper model (blue) and Fenard model (black); $\phi$ = 0.5, 1250 K, 5 bar and 20% fuel consumption. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

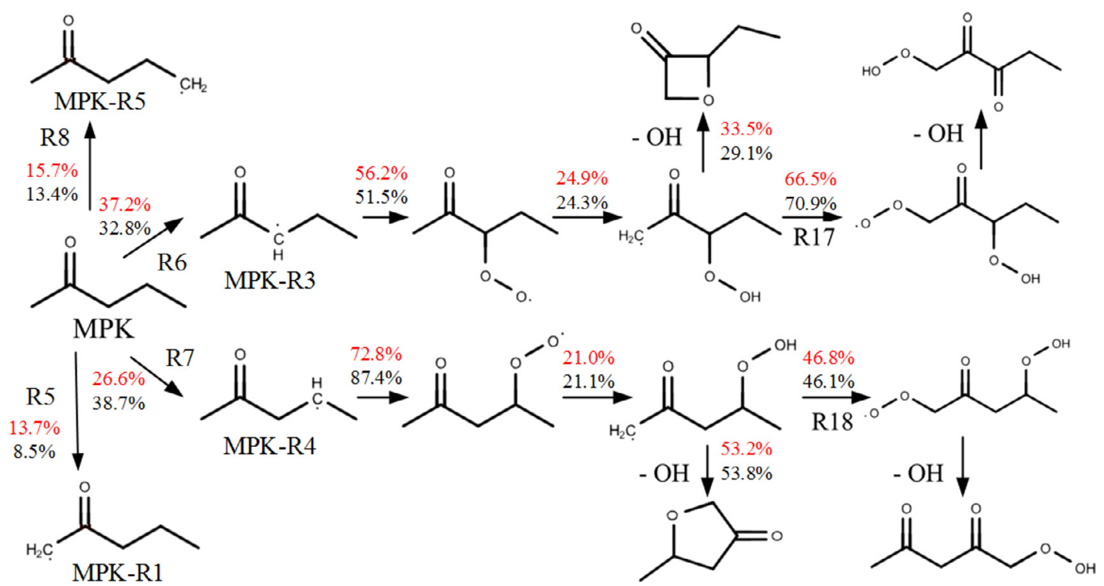


**Fig. 15.** Reaction flux analysis for MPK using the optimized model (red) and Fenard model (black); $\phi$ = 1, 700 K, 20 bar and 20% fuel consumption. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

main source of R18, and 26.6% MPK is consumed by R7 in the optimized model while the corresponding value in Fenard model is 38.7%. From Fig. 15, R7 is overestimated in Fenard model, which results in the enhancement of R18. To this end, the rate constants of R17 and R18 were reduced by a factor of 0.5 in Fenard model to predict well the IDTs measured in RCM. In the optimized model, the rate constants of R17 and R18 are the same as those of the first $O_2$ addition reactions. Therefore, the overestimation of R7 and underestimation of R8, R17, and R18 are the main reason for the overestimation of IDTs by Fenard model.

### 5.4. Model comparisons for laminar flame speed prediction

At the condition of $P$ = 5 atm and $\varphi$ = 1.1, the optimized model well predicts the LFSs within the experimental uncertainty of ±1.2 cm/s [11], while Fenard model and Pieper model underpredict the laminar flame speeds by 3.7 cm/s and 2.0 cm/s respectively. The flow rate sensitivity analysis is performed using the above three models at $P$ = 5 atm and $\varphi$ = 1.1 and the results are shown in Fig. 16. From Fig. 16, the hydrogen abstraction reaction by H, R11 (MPK + H <=> MPK-R4 + $H_2$), shows obvious inhibition effects to the flame propagation in Fenard model, while the fuel-related reactions have almost no influence in the other two models. The rate constants of R11 were decreased by a factor of 0.34 in the optimized model, as shown in Table 3 in Section 4. 5.4% and 15.9% fuel is consumed by R11 in the optimized model and Fenard model, respectively, as shown in Fig. 17. Therefore, the overestimation of R11 is one reason for the underprediction of the flame speeds in Fenard model. In addition, $C_2H_5$ is an important radical to promote flame propagation through the decomposition reaction (R207), as depicted in Fig. 16. The $C_2H_5$ sensitivity analysis shows that the MPK decomposition reaction (R3) makes a significant contribution to the $C_2H_5$ formation. Figure 17 shows that 3.3% and 15.7% fuel is consumed by R3 in Fenard model and the optimized model respectively. Therefore, the underestimation of R3 is another reason for the underprediction of the flame speeds in Fenard model.

As analyzed in Section 5.3, R3 is underestimated in Pieper model. Consequently, 3.3% and 15.7% fuel is consumed by R3 in Pieper model and the optimized model respectively, as depicted in

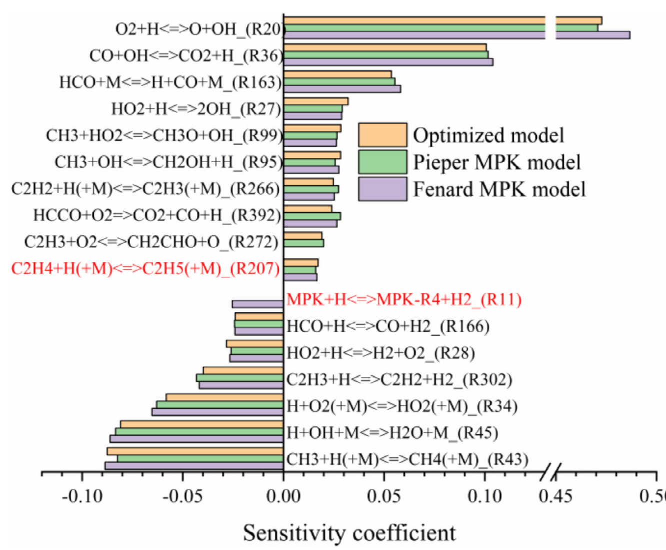


**Fig. 16.** Flow rate sensitivity analysis using the pentanone isomers model at $P$ = 5 atm and $\varphi$ = 1.1.

Fig. 17. As a result, the underestimation of the R3 is the reason for the underprediction of the flame speeds in Pieper model.

### 5.5. Hydrogen abstraction reactions, decomposition reactions and optimization

The flux analysis for MPK using the optimized model, Pieper model, and Fenard model were conducted at $T$ = 1250 K and at $T$ = 700 K respectively, which could be find in Supplementary Materials. The decomposition reactions have evident effects on the MPK oxidation only in high-temperature oxidation while the hydrogen abstraction reactions have significant effects in low and high-temperature oxidation. Obviously, the competition between the decomposition reactions and hydrogen abstraction reactions of MPK is very important to the MPK high temperature oxidation. Figure 18 shows the ratios of MPK consumed by R1 to R4 in the optimized model, Pieper model and Fenard model at $T$ = 1250 K. The ratios of the MPK consumed by the decom-

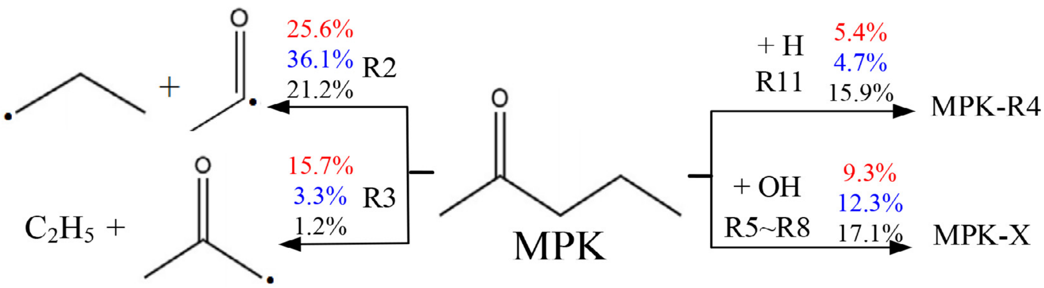


**Fig. 17.** Reaction flux analysis for MPK using the optimized model (red), Pieper model (blue) and Fenard model (black); $\phi$ = 1.1, 5 atm and peak point of heat release. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

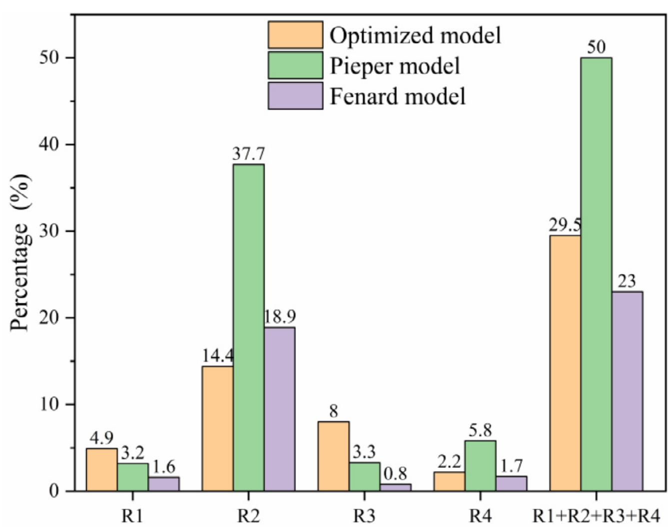


**Fig. 18.** Ratio of MPK consumed by R1 to R4 in the optimized model, Pieper model and Fenard model at $T$ = 1250 K.

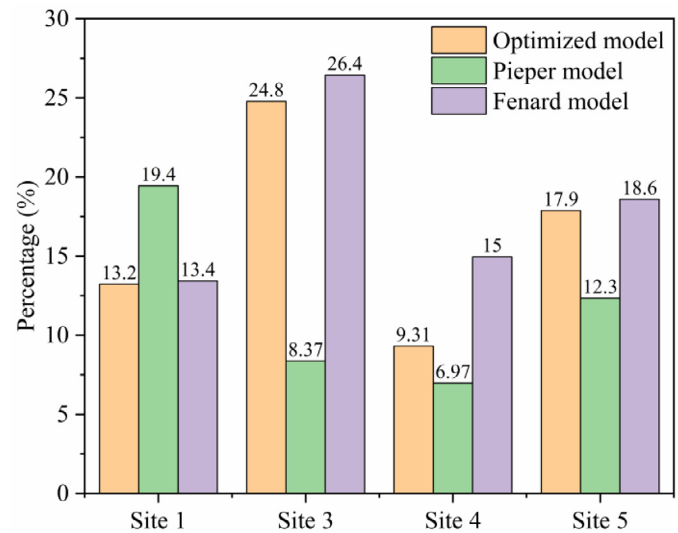


**Fig. 19.** Ratio of MPK consumed by hydrogen abstraction reactions at site 1, 3, 4, and 5 in the optimized model, Pieper model and Fenard model at $T$ = 1250 K.

position reactions and hydrogen abstraction reaction in the optimized model, Pieper model and Fenard model are 29.5%:70.5%, 50.0%:49.5% and 23.0%:76.8% at $T$ = 1250 K, respectively, which indicate that the decomposition reactions are overestimated in Pieper model while underestimated in Fenard model. The rate constants of R1, R2, and R3 were increased by a factor of 4 in Pieper model, and are the same as those of the corresponding reactions of 2-butanone in Fenard model, which indicates that the branching ratio among R1, R2 and R3 in Pieper and Fenard models are the same as that in 2-butanone. Because the third C–C bond in MPK is its typical bond and distinctly different from that in 2-butanone, remaining the branching ratio among R1, R2 and R3 unchanged in MPK is unnecessary.

In respect of the relation of the optimization of the rate constants of the decomposition reactions and BDE, as shown in Table 3 and Fig. 1, the BDEs of the second and fourth C–C bonds of MPK are very close to those of 2-butanone and pentane, respectively, and the rate constants of R2 and R4 in the optimized model are equal or very close to those of the corresponding reactions of 2-butanone and pentane, respectively. The differences in the BDEs of the first and the third C–C bonds between MPK and 2-butanone are −1.3 and −4.3 kcal/mol, respectively, and the pre-exponential factors of R1 and R3 are increased by factors of 3.9 and 6 in the optimized model, respectively. Therefore, the modification of the pre-exponential factors of the decomposition reactions of MPK based on the original pre-exponential factors has a positive relationship with the difference of the BDE of the corresponding C–C bond between MPK and 2-butanone.

Figure 19 shows that the ratios of MPK consumed by the hydrogen abstraction reactions at carbons 1, 3, 4, and 5 in the optimized model, Pieper model, and Fenard model at $T$ = 1250 K. From Fig. 19, the ratios of MPK consumed by the hydrogen abstraction reactions at carbons 1 and 5 in the optimized model are very close to those in Fenard model, while ones at carbons 3 and 4 in the optimized model are less than those in Fenard model. In Fenard model, the rate constants of R5–16 remain unchanged compared to the original rate constants. In the optimized model, the pre-exponential factors of the hydrogen abstraction reactions by OH and H at carbons 1, 3, and 5 are reduced by factors of 0.45 and 0.34, respectively, while ones by $CH_3$ remain unchanged. The pre-exponential factors of the hydrogen abstraction reactions by OH, H, and $CH_3$ at carbon 4 are reduced by factors of 0.34, 0.33, and 1.0 based on pentane, respectively. These indicate that the differences between the MPK molecular and 2-butanone have a larger impact on the H-abstraction reactions by OH and H radicals than the H-abstraction reactions by $CH_3$. Therefore, the ratio of the rate constants among the hydrogen abstraction reactions by OH, H, and $CH_3$ is different from that of the original rate constants while the branching ratio of the hydrogen abstraction reactions at carbon 1, 3, and 5 remains unchanged.

In the low temperature oxidation, the hydrogen abstraction reactions dominate the consumption of MPK, and the overwhelming majority of MPK is consumed by the hydrogen abstraction reactions by OH. The reduction in the pre-exponential factors of the hydrogen abstraction reactions by OH at carbon 1, 3, and 5 (R5, R6, and R8) in the optimized MPK model is less than that at carbon 4

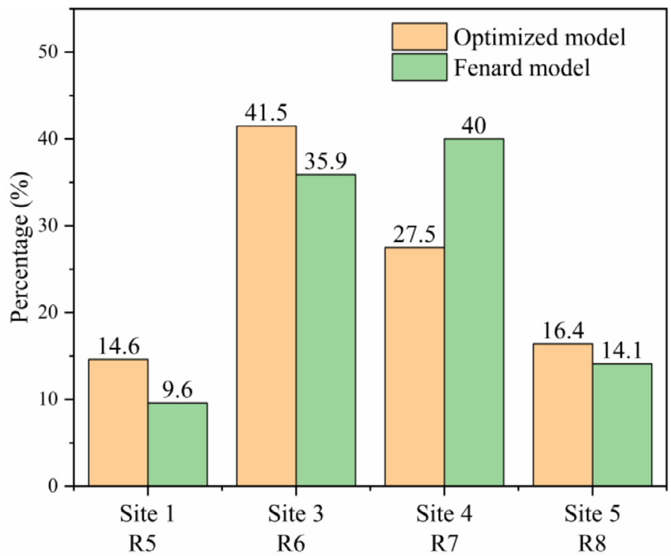


**Fig. 20.** Ratio of MPK consumed by hydrogen abstraction reactions by OH at site 1, 3, 4, and 5 in the optimized model and Fenard model at $T = 700$ K.

(R7), which leads to the reduction in the ratio of MPK consumed by R7 and increase in the ratios of MPK consumed by R5, R6 and R8 in the optimized model compared to Fenard model, as depicted in Fig. 20. Obviously, the hydrogen abstraction reactions at carbon (1, 3 and 5) and at carbon 4 are underestimated and overestimated in Fenard model, respectively.

In summary, the competition among the decomposition and hydrogen abstraction reactions of MPK are intricate and crucial to the low and high-temperature oxidation. It is necessary to globally optimize the rate constants of R1–R16 based on the multidimensional experimental results, such as low and high-temperature IDTs, LFSs, and species profiles.

## 6. Conclusion

The IDTs of MPK were measured at equivalence ratios of 0.5, 1.0, and 1.5, pressures of 1 bar and 5 bar, and temperatures ranging between 1227 and 1571 K. A MPK low to high-temperature kinetic was constructed on the basis of Pieper model and Fenard model. The modification of the model is the optimization of the rate constants of R1–R16 obtained by the analogy-based method by Bayesian Optimization algorithm using the high-temperature IDTs. The optimized model predicts well the LFSs (measured by Li et al.) and the IDTs and species profiles in low temperature (measured by Fenard et al.). There is no overfitting during the optimization process. Therefore, the optimization method is reliable.

Because the third C–C bond in MPK is its typical bond and distinctly different from that in 2-butanone, remaining the branching ratio among the three decompositions (R1–R3) and the ratio among the rate constants of the hydrogen abstraction reactions by OH, H, and $CH_3$ unchanged in MPK is unnecessary. The modification of the rate constants of the decomposition reactions of MPK based on the original rate constants has a positive relationship with the difference of the BDE of the corresponding C–C bond between MPK and 2-butanone. It is necessary to globally optimize the rate constants of R1–R16 based on the multidimensional experimental results, such as low and high-temperature IDTs, LFSs, and species profiles.

## Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.111453.

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